

Dyumna Negi

9068656899 | dyumna.negi1@gmail.com | [linkedin.com/dyumna-negi](https://www.linkedin.com/in/dyumna-negi) | github.com/dyumna11

EDUCATION

Indian Institute of Information Technology

B.Tech Electronics and Communication Engineering (CGPA-8.6)

Jabalpur, MP

Aug. 2024 – May 2028

St. Joseph's Academy

ISC - 94.75% (Top 3% of batch)

Dehradun, UK

Aug. 2023 – May 2024

St. Joseph's Academy

ICSE - 98.8% (City Rank 11, 3rd School Topper)

Dehradun, UK

Aug. 2021 – May 2022

PROJECTS

ParkWise– AI-Driven Parking Intelligence & Enforcement System | Python, Pandas, LightGBM, NetworkX [GitHub](#)

- Built an AI-powered parking analytics platform using Python, Pandas, and NetworkX to analyze 25K+ parking violation records and identify congestion hotspots.
- Created a LightGBM forecasting pipeline with Mann-Kendall trend analysis to predict emerging hotspots.
- Engineered a Parking Congestion Impact Index (PCII) and implemented Granger Causality-based congestion propagation analysis.

Medical Reminder App | React.js, Node.js, Express.js, MongoDB, JWT

[GitHub](#) | [Live Demo](#)

- Built and deployed a cloud-hosted Medical Reminder application with 5+ core features, enabling secure medication scheduling, tracking, and automated notifications.
- Implemented secure authentication using JWT and bcrypt, protecting user credentials through password hashing and token-based authorization.
- Developed an automated reminder workflow using Node-Cron and Resend, executing reminder evaluations every minute and delivering real-time email notifications.

Traffic Demand Forecasting for Smart Signal Systems | Python, CatBoost, Pandas, NumPy

[GitHub](#)

- Extracted 15+ temporal, geospatial, and road-network features from traffic data, including cyclical time encoding and hierarchical geohash representations.
- Trained and evaluated an ensemble of CatBoost regression models to forecast traffic demand across urban road segments.
- Attained an R^2 score of approximately 0.95 through feature engineering, hyperparameter tuning, and ensemble learning techniques.

VCG Signal Compression and Reconstruction using Matrix Completion | Python, NumPy, SciPy, Matplotlib, Git

- Proposed a biomedical signal compression framework using matrix completion and low-rank recovery techniques for VCG reconstruction.
- Implemented subsampling and reconstruction algorithms, achieving a Compression Ratio (CR) of 30 while maintaining a Percentage Root Difference (PRD) of 5.
- Evaluated reconstruction quality across multiple datasets; research manuscript currently under preparation based on the proposed methodology.

SKILLS

Programming: C++, Java, Python, JavaScript

Machine Learning: Scikit-Learn, Pandas, NumPy, Matplotlib, Data Preprocessing, Feature Engineering, Model Evaluation, Classification

Web: React.js, Node.js, Express.js, HTML, CSS, REST APIs

Database: MongoDB, MongoDB Atlas

Tools: Git, GitHub, Postman, VS Code

Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Probability & Statistics, Linear Algebra

Areas of Interest: Machine Learning, Signal Processing, Computer Vision, Data Analytics

ACHIEVEMENTS

- Selected as a **Mentee in the LinkedIn CoachIn Program** through a highly competitive national selection process (**21,000+ applicants**).
- Advanced to Round 2 of **Flipkart GRID**, placing in the **top 17%** of 10,000+ teams by developing a machine learning-based traffic prediction model.
- Chosen as a **Mentee at Codess Cafe** via a selective screening process (**800+ applicants**).
- Solved **850+ algorithmic problems** across [LeetCode](#) (**Rating: 1670**), [Codeforces](#), [GeeksforGeeks](#), and [CodeChef](#) (**2-Star**).
- Secured a spot in the **Top 25 teams** of **CodeRumble 3.0**, an ICPC-style contest with around 250 participants.
- Earned **Elite + Gold Certification** in **Internet of Things** and **Programming in Java** through NPTEL-SWAYAM.